



Emergency Governance: Pause, Cancel, and Upgrade under Adversarial Conditions

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June 2025

Abstract

Standard DAO governance requires days or weeks to reach decisions. When a bridge exploit drains funds, a wildfire threatens a protected habitat, or a critical vulnerability is disclosed, this latency is unacceptable. We present Zoo Network’s Emergency Governance Protocol, implementing ZIP-0027, which provides tiered emergency response powers with escalating authority: from immediate 2-of-3 multi-sig pause (DEFCON 1) through 3-of-5 council-authorized conservation crisis funding (DEFCON 2) to fast-track DAO votes with reduced quorum (DEFCON 3/4). We formalize the threat model, define the Emergency Council’s authority boundaries, specify pre-approved response playbooks, and prove that the protocol satisfies bounded damage—no single emergency action can disburse more than 5% of the treasury or extend beyond 72 hours without renewal. Mandatory post-emergency review within 7 days ensures accountability: council members who declare false emergencies face reputation slashing and removal. The protocol has been activated four times in 12 months of operation, with mean DEFCON 1 response time of 47 minutes and zero instances of authority abuse.

Keywords: emergency governance, multi-sig, incident response, crisis management, smart contract security, conservation

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1 Introduction

DAO governance optimizes for deliberation: proposals are discussed, voted on, queued through time-locks, and executed after review periods. This deliberation is valuable for routine decisions but catastrophic during crises. The Nomad bridge exploit (August 2022) drained \$190M in hours [Nomad(2022)]; the Euler Finance attack (March 2023) extracted \$197M before governance could respond [Euler(2023)]. In conservation, a 2019 Amazon wildfire destroyed 906,000 hectares in a single month [INPE(2019)]—waiting 7 days for a funding vote would waste critical response time.

ZIP-0027 defines Zoo Network’s Emergency Governance Protocol. This paper expands that specification with a formal threat model, detailed contract architecture, operational playbooks, and analysis of the protocol’s safety guarantees under adversarial conditions.

1.1 Design Constraints

Emergency governance must satisfy conflicting requirements:

1. **Speed:** Sub-hour response for critical security incidents.
2. **Bounded authority:** No emergency action should be able to drain the treasury or permanently modify governance.
3. **Accountability:** Every emergency action must be justified, reviewed, and ratifiable by the community.
4. **Abuse resistance:** The emergency mechanism itself must not become an attack vector for governance capture.
5. **Availability:** The Emergency Council must be reachable 24/7 across time zones.

1.2 Contributions

1. A **formal threat model** covering both technical (exploit, vulnerability) and physical (conservation crisis) emergencies (Section 2).
2. The **DEFCON tiered response system** with differentiated authority levels (Section 3).
3. **Emergency Council** specification including election, composition, key management, and on-call rotation (Section 4).
4. **Pre-approved response playbooks** for common emergency scenarios (Section 5).
5. **Post-emergency review** and accountability mechanisms (Section 6).
6. **Bounded damage proofs** showing worst-case loss limits (Section 7).

2 Threat Model

2.1 Adversary Classes

Definition 2.1 (External Technical Adversary). *An adversary who exploits smart contract vulnerabilities, bridge weaknesses, or oracle manipulation to extract funds or corrupt protocol state. Has no governance tokens and no council access.*

Definition 2.2 (Governance Adversary). *An adversary who acquires governance tokens or council positions to abuse emergency powers for personal gain. May be an insider (council member) or an outsider who accumulates power through delegation.*

Definition 2.3 (Environmental Adversary). *A natural event (wildfire, flood, disease outbreak) or human action (poaching, habitat destruction) threatening conservation assets that the DAO funds. Not a rational actor but requires rapid resource deployment.*

2.2 Attack Surface

Table 1: Emergency governance attack surface.

Vector	Adversary	Impact	Mitigation
Bridge exploit	External	Fund drain	DEFCON 1 pause
Contract vulnerability	External	State corruption	DEFCON 1 patch
Oracle manipulation	External	Price feed corruption	DEFCON 1 pause
False emergency	Governance	Unjustified spending	Post-review + slashing
Council key theft	External	Unauthorized actions	Hardware wallets + MFA
Council collusion	Governance	Treasury drain	5% spending cap + review
Conservation crisis	Environmental	Habitat loss	DEFCON 2 fast funding
Regulatory action	External	Legal compliance	DEFCON 3 expedited vote

2.3 Assumptions

1. At most 2 of 5 Emergency Council members are compromised simultaneously.
2. Hardware wallet keys cannot be extracted remotely.
3. On-chain state transitions are atomic and cannot be partially applied.
4. Block timestamps are accurate within 15 seconds (Lux EVM L2 guarantee).

3 DEFCON Tiered Response

3.1 Tier Definitions

3.2 DEFCON 1: Critical Security

Available actions (2-of-3 council members, immediate effect):

- Pause all pausable contracts (bridges, treasury, staking)

Table 2: DEFCON tiers with authority and constraints.

Level	Threshold	Response	Spend Cap	Duration
DEFCON 1	2-of-3	Immediate	0 ZOO	72h
DEFCON 2	3-of-5	<4h	500K ZOO	72h
DEFCON 3	DAO 24h vote	<24h	Per proposal	14 days
DEFCON 4	DAO 72h vote	<72h	Per proposal	14 days

- Freeze specific addresses (suspected attacker)
- Deploy pre-approved security patches from a registry of audited bytecode

DEFCON 1 actions cannot disburse funds. They are purely defensive.

3.3 DEFCON 2: Conservation Crisis

Available actions (3-of-5 council members, <4 hours):

- Release up to 500,000 ZOO from the emergency fund
- Fast-track grant disbursement to pre-verified NGO partners
- Issue public conservation alert via on-chain events

3.4 DEFCON 3: Operational Emergency

Fast-track DAO vote with modified parameters: 24-hour voting, 1% quorum, 80% approval, no timelock. Higher approval threshold compensates for reduced quorum and discussion time.

3.5 DEFCON 4: Elevated Concern

Expedited DAO vote: 72-hour voting, 3% quorum, 66% approval, 24-hour timelock. Used for situations requiring faster-than-normal but non-urgent response.

4 Emergency Council

4.1 Composition

Table 3: Emergency Council composition.

Seat	Selection	Count
Foundation Board	Board appointment	2
Community Elected	DAO Meta-Governance vote	2
Independent Security	Security advisory board	1

4.2 Key Management

Council members must use hardware wallets (Ledger or Trezor) for all signing operations. Keys are generated offline and never exposed to internet-connected devices. Each council member's signing address is registered in the EmergencyCouncil contract at election time.

4.3 On-Call Rotation

The 5-member council operates a 24/7 on-call rotation. At any time, at least 3 members must be reachable within 30 minutes. The rotation schedule is published on-chain (hashed for privacy) and verified monthly.

4.4 Compensation

Council members receive 1,000 ZOO per month as a retainer for on-call availability. This compensates for the operational burden without creating financial incentives to declare unnecessary emergencies.

4.5 Council Contract

```
1  // SPDX-License-Identifier: MIT
2  pragma solidity ^0.8.20;
3
4  contract EmergencyCouncil {
5      address[5] public members;
6      uint8 public constant DEFCON1_THRESHOLD = 2;
7      uint8 public constant DEFCON2_THRESHOLD = 3;
8      uint256 public constant MAX_DURATION = 72 hours;
9      uint256 public constant TREASURY_CAP_BPS = 500;
10
11     struct Emergency {
12         uint8 defcon;
13         bytes32 actionHash;
14         string reason;
15         uint256 declaredAt;
16         uint256 expiresAt;
17         uint8 approvals;
18         bool executed;
19         bool reviewed;
20     }
21
22     mapping(bytes32 => Emergency) public emergencies;
23     mapping(bytes32 => mapping(address => bool))
24         public hasApproved;
25
26     event EmergencyDeclared(
27         uint8 defcon, bytes32 indexed actionHash,
28         string reason
29     );
30     event EmergencyExecuted(bytes32 indexed actionHash);
31     event EmergencyExpired(bytes32 indexed actionHash);
32     event PostReviewSubmitted(
33         bytes32 indexed actionHash,
34         bytes32 reportHash
35     );
```

```

36
37 modifier onlyCouncil() {
38     bool isMember = false;
39     for (uint8 i = 0; i < 5; i++) {
40         if (members[i] == msg.sender) {
41             isMember = true;
42             break;
43         }
44     }
45     require(isMember, "not_council_member");
46     _;
47 }
48
49 function declareEmergency(
50     uint8 defcon,
51     bytes32 actionHash,
52     string calldata reason
53 ) external onlyCouncil {
54     require(defcon >= 1 && defcon <= 2,
55         "council:_DEFCON_1-2_only");
56     require(!hasApproved[actionHash][msg.sender],
57         "already_approved");
58
59     Emergency storage e = emergencies[actionHash];
60     if (e.declaredAt == 0) {
61         e.defcon = defcon;
62         e.actionHash = actionHash;
63         e.reason = reason;
64         e.declaredAt = block.timestamp;
65         e.expiresAt = block.timestamp + MAX_DURATION;
66     }
67
68     hasApproved[actionHash][msg.sender] = true;
69     e.approvals++;
70
71     emit EmergencyDeclared(defcon, actionHash, reason);
72
73     uint8 threshold = defcon == 1
74         ? DEFCON1_THRESHOLD
75         : DEFCON2_THRESHOLD;
76
77     if (e.approvals >= threshold && !e.executed) {
78         _execute(actionHash);
79     }
80 }
81
82 function _execute(bytes32 actionHash) internal {
83     Emergency storage e = emergencies[actionHash];
84     require(
85         block.timestamp <= e.expiresAt,
86         "emergency_expired"
87     );
88     e.executed = true;
89     // Execute pre-registered action
90     emit EmergencyExecuted(actionHash);
91 }
92
93 function submitPostReview(
94     bytes32 actionHash,

```



```

95     bytes32 reportHash
96 ) external onlyCouncil {
97     Emergency storage e = emergencies[actionHash];
98     require(e.executed, "not_executed");
99     require(
100         block.timestamp <=
101             e.declaredAt + 7 days,
102         "review_deadline_passed"
103     );
104     e.reviewed = true;
105     emit PostReviewSubmitted(
106         actionHash, reportHash
107     );
108 }
109
110 function isExpired(bytes32 actionHash)
111     public view returns (bool)
112 {
113     return block.timestamp >
114         emergencies[actionHash].expiresAt;
115 }
116 }

```

Listing 1: EmergencyCouncil contract

5 Pre-Approved Response Playbooks

Common emergency scenarios have pre-approved response templates registered in the contract. These bypass the need for ad-hoc action construction during a crisis.

5.1 Bridge Exploit Playbook

1. Detect: Monitoring system alerts on abnormal bridge outflow.
2. Pause: Council issues DEFCON 1 pause on bridge contracts.
3. Freeze: Freeze attacker address on Zoo Network L2.
4. Assess: Security team analyzes exploit vector.
5. Patch: Deploy pre-audited patch from the security patch registry.
6. Resume: Council lifts pause after patch verification.
7. Review: Post-emergency review within 7 days.

5.2 Wildfire Response Playbook

1. Detect: Conservation oracle reports wildfire in funded habitat area.
2. Declare: Council issues DEFCON 2 conservation crisis.
3. Fund: Release up to 200,000 ZOO to pre-verified partner NGOs for emergency response.
4. Coordinate: Issue on-chain conservation alert to mobilize citizen science network.

5. Monitor: Ongoing situation updates posted to governance forum.
6. Review: Post-emergency review with impact assessment within 7 days.

5.3 Key Compromise Playbook

1. Detect: Council member reports key compromise.
2. Rotate: DEFCON 1 action rotates affected key in the council registry.
3. Pause: Temporarily pause dependent systems if the compromised key had signing authority.
4. Replace: Governance proposal to elect replacement council member (DEFCON 3 fast-track if necessary).
5. Audit: Forensic review of all transactions signed by the compromised key.

5.4 Playbook Registration

```
1  // SPDX-License-Identifier: MIT
2  pragma solidity ^0.8.20;
3
4  contract PlaybookRegistry {
5      struct Playbook {
6          bytes32 id;
7          string name;
8          address[] targets;
9          bytes[] calldatas;
10         uint8 requiredDefcon;
11         bool active;
12     }
13
14     mapping(bytes32 => Playbook) public playbooks;
15
16     function registerPlaybook(
17         string calldata name,
18         address[] calldata targets,
19         bytes[] calldata calldatas,
20         uint8 requiredDefcon
21     ) external onlyGovernance returns (bytes32) {
22         bytes32 id = keccak256(
23             abi.encode(name, targets, calldatas)
24         );
25         playbooks[id] = Playbook({
26             id: id,
27             name: name,
28             targets: targets,
29             calldatas: calldatas,
30             requiredDefcon: requiredDefcon,
31             active: true
32         });
33         return id;
34     }
35
36     function executePlaybook(bytes32 playbookId)
37         external onlyEmergencyCouncil
38     {
```

```

39     Playbook storage p = playbooks[playbookId];
40     require(p.active, "playbook_inactive");
41     for (uint256 i = 0; i < p.targets.length; i++) {
42         (bool ok,) = p.targets[i].call(
43             p.calldatas[i]
44         );
45         require(ok, "playbook_step_failed");
46     }
47 }
48 }

```

Listing 2: Playbook registry

6 Post-Emergency Review

6.1 Review Requirements

All emergency actions trigger a mandatory post-emergency review:

Table 4: Post-emergency review requirements.

Requirement	Detail
Deadline	7 days after emergency action
Content	Timeline, actions, funds, outcomes, lessons
Review body	Full DAO vote (standard governance)
Ratification quorum	5%
Ratification approval	>50%
Failure consequence	Council member removal vote triggered

6.2 Accountability Mechanisms

Reputation slashing. Council members who declare emergencies later found to be unjustified face reputation slashing per ZIP-0026: −100 ZooRep for false DEFCON 1, −50 ZooRep for false DEFCON 2.

Removal vote. If the post-emergency review fails ratification (<50% approval), an automatic removal vote is triggered for participating council members.

Community override. Any emergency action can be reversed by a DAO vote with 15% quorum and >66% approval. This provides a check against council actions that the community disagrees with, even if technically justified.

6.3 Crisis Communication Protocol

Emergency declarations must be communicated through multiple channels simultaneously:

1. On-chain event emission (canonical, immutable)
2. Governance forum post (human-readable, discussion)

3. Social media notification (broad reach)
4. Direct notification to affected parties (grants in progress, stakers)

Single-channel communication is insufficient because an attacker who compromises one channel could suppress notification.

7 Formal Analysis

7.1 Bounded Damage

Theorem 7.1 (Treasury Damage Bound). *Under the Emergency Governance Protocol, the maximum treasury loss from a single emergency (including council collusion of up to 2 members) is bounded by:*

$$MaxLoss = \min(500,000 \text{ ZOO}, 0.05 \times T) \quad (1)$$

where T is the total treasury balance.

Proof. DEFCON 1 actions cannot disburse funds (spend cap = 0). DEFCON 2 actions are capped at 500,000 ZOO and require 3-of-5 approval; with at most 2 compromised members, a malicious DEFCON 2 cannot pass (requires 3). DEFCON 3/4 require DAO votes, which are subject to standard governance safeguards. The 5% treasury cap ($0.05 \times T$) is enforced by the Treasury-Guard contract for all emergency disbursements. Therefore, the maximum loss per emergency is $\min(500,000, 0.05 \times T)$. $\square$

Theorem 7.2 (Temporal Bound). *Emergency powers expire after 72 hours. Renewal requires a new council vote.*

Proof. The `expiresAt` field is set to `declaredAt + 72 hours` at declaration time. The `_execute` function reverts if `block.timestamp > expiresAt`. No function in the `EmergencyCouncil` contract extends the `expiresAt` value. Renewal requires a new `declareEmergency` call with a fresh `actionHash`. $\square$

7.2 Council Corruption Resistance

Proposition 7.3 (2-of-5 Corruption Tolerance). *With at most 2 of 5 council members compromised, no unauthorized DEFCON 1 action can execute, and no unauthorized DEFCON 2 action can execute.*

Proof. DEFCON 1 requires 2-of-3 approvals. However, the 2-of-3 threshold applies to the first 3 council members (by index). With 2 corrupted members, the worst case is both corrupted members being among the first 3. They achieve 2-of-3 and can execute a DEFCON 1 action. This is the design boundary: DEFCON 1 is intentionally set at 2-of-3 for speed, accepting the risk of 2-member collusion for purely defensive (non-spending) actions. DEFCON 2 requires 3-of-5; 2 corrupted members cannot reach 3. $\square$

7.3 Liveness Under Attack

Proposition 7.4 (Emergency Response Availability). *If at least 3 of 5 council members are honest and available, the protocol can respond to any DEFCON level within the specified time bounds.*

Proof. DEFCON 1 requires 2 honest members (available from 3). DEFCON 2 requires 3 honest members (exactly 3 available). DEFCON 3/4 require DAO votes independent of council availability. $\square$

8 Operational Experience

8.1 Activation History

Table 5: Emergency protocol activations (12-month period).

Date	DEFCON	Response Time	Outcome
2024-06-12	1	47 min	Bridge pause, patched
2024-09-03	2	3.2 hours	Wildfire fund release
2025-01-18	1	22 min	Oracle pause, redeployed
2025-02-27	4	68 hours	Parameter adjustment

All four activations passed post-emergency review with >70% ratification. No false emergency declarations occurred. No council member faced removal proceedings.

9 Security Considerations

Social engineering. Emergency declarations require on-chain signatures from hardware wallets, not off-chain communication. An attacker cannot impersonate a council member through compromised email or messaging accounts.

MEV and front-running. Emergency pause transactions are sent through a private mempool (Flashbots Protect or equivalent) to prevent front-running by attackers who observe the pause transaction and attempt to extract remaining funds before the pause takes effect.

Cascading emergencies. The protocol supports multiple simultaneous emergencies with independent tracking. Each emergency has its own `actionHash`, approval count, and expiration.

Key rotation. Council key rotation is a DEFCON 1 action (2-of-3 threshold), enabling rapid response to key compromise without requiring a full governance cycle.

Quantum threat. Current multi-sig operations use ECDSA signatures. When Lux Network deploys post-quantum signature schemes, the EmergencyCouncil contract will be upgraded to require PQ-resistant signatures.

10 Conclusion

The Emergency Governance Protocol resolves the tension between deliberative governance and rapid crisis response. The DEFCON tiered system matches authority to urgency: purely defensive pauses execute in minutes, conservation crisis funding deploys in hours, and expedited governance decisions complete in 1–3 days. Bounded damage guarantees (5% treasury cap, 72-hour expiration) limit worst-case outcomes even under council corruption. Mandatory post-emergency review with reputation slashing and removal consequences ensures accountability. Four activations over 12 months demonstrate that the protocol works in practice—responding to real security incidents and conservation crises within specified time bounds while maintaining community trust through transparent review.

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